Low-Level Sensor Signal Conditioning Circuit Design

CIE 201 — Advanced Electric Circuits | Phase 1 Report | Spring 2026

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Abstract — *This report presents the first phase of the CIE 201 project on low-level sensor signal conditioning. The selected sensor is a K-Type thermocouple, which produces a raw analog voltage in the microvolt range. The proposed circuit conditions this signal through four stages: instrumentation amplification, low-pass filtering, output scaling, delivering a clean 0 V to 2 V output. This document covers sensor selection, datasheet analysis, and the complete circuit block diagram.*

1. TEAM MEMBER DUTIES

Name	Student ID	Duties
Mohammed Soliman	202402280	Sensor Selection & Analysis & Protus Design
Mohamed Tamer	202400792	Hardware Circuit Implementation & Hardware Circuit Testing and Verification
Omar Ihab Fared Abdo	202401437	Block diagram design & documentation & Simulation of the output signals (Hardware)
Ahmed Akef	202401486	Signal analysis & waveform verification & Calculations
Karim Assem	202400688	Report writing & Hardware Circuit Testing and Verification & simulation

2. CHOSEN SENSOR

The selected sensor for this project is the **K-Type Thermocouple** (Chromel–Alumel pair), used in its bare wire form without any digital interface module. The thermocouple operates on the Seebeck effect, generating a small voltage proportional to the temperature difference between its two junctions. This raw voltage lies in the microvolt to millivolt range, making it an ideal low-level AC signal source for this project.

The sensor was selected for the following reasons:

a) It produces a genuine low-level analog signal in the µV–mV range, fully satisfying the project's input requirements.

b) The thermocouple wires naturally pick up 50 Hz electromagnetic interference from the power grid, making noise filtering both necessary and clearly demonstrable.

c) With a sensitivity of 41 µV/°C and a range up to 1024°C, it provides a well-characterized signal suitable for circuit design.

d) It is widely used in industrial temperature monitoring, giving this project real-world relevance.

e) The differential output is ideal for demonstrating the operation of an instrumentation amplifier.

3. DATASHEET ANALYSIS

3.1 Output Characteristics

Based on the MAX6675 datasheet (Maxim Integrated, Rev. 3, June 2021), the K-Type thermocouple output voltage follows the equation:

$$\boldsymbol{V}_{\mathrm{OUT}} = \mathbf{41}\ \boldsymbol{\mu}\mathbf{V}/^{\circ}\mathbf{C}\cdot(\boldsymbol{T}_{\mathrm{remote}} - \boldsymbol{T}_{\mathrm{ambient}})$$

The sensor sensitivity is **41 µV per °C**, with a measurement range from 0°C to 1024°C at the hot junction. The cold junction (ambient side) is rated for −20°C to +85°C. The raw output voltage ranges from approximately 0 µV at 0°C to about 50 mV at 1250°C.

3.2 Frequency Response and Noise

The thermocouple signal is inherently quasi-DC — temperature changes are very slow, well below 1 Hz. However, the thermocouple wires act as antennas and pick up **50 Hz electromagnetic interference** from Egypt's power grid, with a typical noise amplitude of approximately 0.3 mV. This noise amplitude can exceed the sensor signal in harsh environments, which is the core motivation for our signal conditioning design.

To reject this interference, our circuit implements a low-pass filter with a cutoff frequency of **fc ≈ 1.6 Hz**, well below the 50 Hz noise frequency. This ensures complete noise attenuation while preserving the temperature information.

3.3 Datasheet Reference

Full datasheet: [MAX6675 Cold-Junction-Compensated K-Thermocouple-to-Digital Converter — Maxim Integrated]

4. CIRCUIT BLOCK DIAGRAM

4.1 System Overview

The signal conditioning circuit takes the raw thermocouple output and processes it through four sequential stages to produce a clean, scaled output between 0 V and 5 V. The signal flow is as follows:

Thermocouple → Instrumentation Amplifier → Low-Pass Filter → Output Scaling

4.2 Stage Descriptions

Stage 1 — K-Type Thermocouple: The bare thermocouple wires generate a differential voltage of ~41 µV/°C superimposed with 50 Hz noise. The raw signal lies between 0 µV and 50 mV depending on temperature.

Stage 2 — Instrumentation Amplifier (MCP6002 × 2): Two MCP6002 op-amps are configured as an instrumentation amplifier. The differential gain is set to 2001× using resistors R3 = R4 = 1MΩ and R5 = 1 kΩ:

$$A_v = 1 + \frac{2R_3}{R_5} = 1 + \frac{2 \times 1\text{M}\Omega}{1\ \text{k}\Omega} = 2001$$

$$V_{\text{out}} = 1\ \text{mV} \times 2001 \approx 2\ \text{V}$$

This stage amplifies the sensor signal from the µV range to approximately 2 V while rejecting common-mode noise through its high CMRR.

Stage 3 — 2nd-Order Low-Pass Filter (R1-C1, R2-C2): A cascaded two-stage RC filter removes the 50 Hz interference. The first stage uses R1 = 1 kΩ and C1 = 100 µF:

$$f_c = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi \cdot 1\ \text{k}\Omega \cdot 100\ \mu\text{F}} \approx 1.6\ \text{Hz}$$

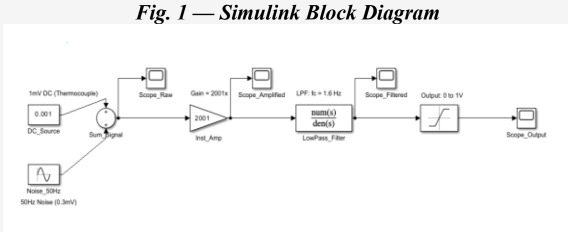
The second stage (R2 = 10 kΩ, C2 = 1000 µF) provides further attenuation with fc2 ≈ 0.016 Hz, ensuring complete noise rejection.

Stage 4 — Output: The filtered signal is passed directly to the output, delivering approximately 2 V DC

4.3 Block Diagram

Figure 1 shows the complete block diagram of the signal conditioning circuit implemented in MATLAB/Simulink.

Fig. 1 — Simulink Block Diagram



4.4 Simulation Results

Figures 2–5 show the oscilloscope readings at each stage of the circuit, confirming correct operation of each block.

Fig. 2 — Scope_Raw: Raw thermocouple signal (1 mV DC + 50 Hz noise)

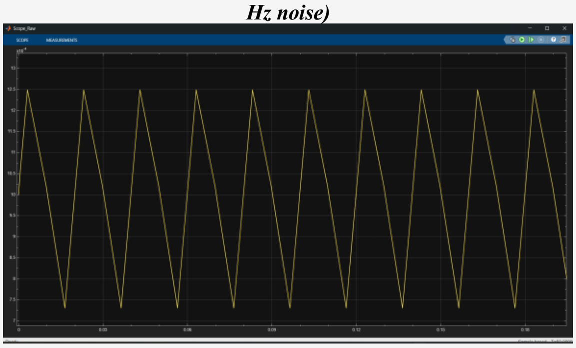


Fig. 3 — Scope_Amplified: Signal after instrumentation amplifier (×2001)

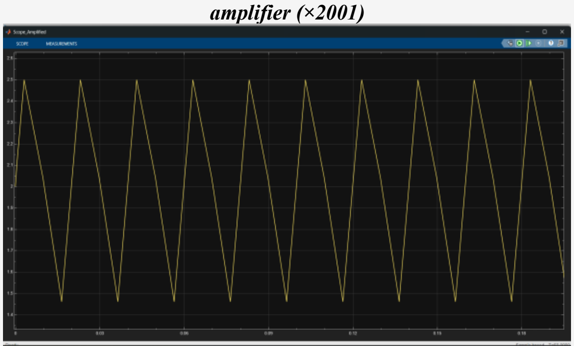


Fig. 4 — Scope_Filtered: Signal after low-pass filter (noise removed)

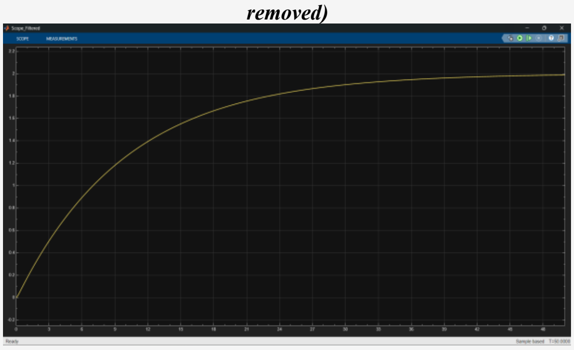


Fig. 5 — Scope_Output: Final output (0 V to 2 V)